Catabolic reactions furnish the energy needed to drive anabolic reactions. This coupling of energy requiring and energy releasing reactions is made possible through the molecule Adenosine tri-phosphate (ATP).

4.2 Energy of Chemical Reaction

Light energy is trapped by phototrophs during photosynthesis, in which it is absorbed by bacteriochlorophyll and other pigments and converted to chemical energy for cellular work. The energy is required by the bacterium for synthesis of cell wall or membrane, synthesis of enzymes, cellular components repair mechanism growth and reproduction.

Some change of energy occurs whenever bonds between atoms are formed or broken during chemical reactions. When a chemical bond is formed, energy is required. Such a chemical reaction which requires energy is called an endergonic reaction, meaning that energy is directed inward. When a bond is broken, energy is released. A chemical reaction that release energy is an exergonic reaction, meaning that energy is directed outward.

During chemical reaction energy is either released or absorbed and the quantum of energy liberated or taken up is that is useful energy and is referred to Free Energy Change (ΔG) of the reactions.

4.2.1 High Energy Phosphate

Adenosine Tri-Phosphate (ATP) is the principal energy carrying molecule of all cells and is indispensable to the life of the cell. It stores the chemical energy released by some chemical reactions, and it provides the energy for reactions that require energy. ATP consists of an adenosine unit composed of adenine, ribose with three phosphate groups. In ATP and some other phosphorylated compounds, the outer two phosphate groups are joined by an **anhydride bond**.

Some of the other high energy nucleotides involved in biochemical processes are given in Table 4.1.

Table 4.1: High energy nucleotides involved in biosynthesis

Name of the Nucleotide	Biosynthesis
Uridine triphosphate (UTP)	Polysaccharide
Cytidine triphosphate (CTP)	Lipid
Guanidine triphosphate (GTP)	Protein

Nutrients are broken from highly reduced compounds to highly oxidized compounds within the cells. Much of the energy released during oxidation - reduction reactions is trapped within the cell by the formation of ATP. A phosphate group is added to ADP with the input of energy to form ATP.

$$ATP + H_2O \longrightarrow ADP + pi \ (\Delta G^\circ = -7.3 \text{ K Cal/mol})$$

$$ATP + H_2O \longrightarrow AMP + ppi \ (\Delta G^\circ = -10.9 \text{ K Cal/mol})$$

ATP is ideally suited for its role as an energy currency. It is formed in energy trapping and energy generating processes such as photosynthesis, fermentation, and aerobic respiration. In bacterial and archeal cells, most of the ATP is formed on the cell membrane, while in eukaryotes the reactions occur primarily in the mitochondria (Figure 4.2).

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